

Project definition — Members & Roles, Objectives and Milestones

Group 12DP-D (Ferran Sánchez · Luz Hurtado · Noa Galván · Leyre Zazpe) · Introduction to Bioinformatics 2026–2027 (UPC)

Project dates: 17/09/2026 – 05/11/2026 (final delivery 26/10/2026; oral presentation 05/11/2026) · 36 working days

1. Project name	
Name	Molecular Analysis of SOD1 Gene Mutations in Amyotrophic Lateral Sclerosis 1 (ALS1)

2. Objectives					
General	Explain how changes in SOD1 gene and protein sequence and structure underlie the molecular basis of ALS1, and what the comparison of SOD1 between humans and other species reveals about the affected biological function.				
ID	Specific objective	Measurable target	How it is verified	Deadline	Work package
SO1	Characterise the molecular cause of ALS1 at gene, protein, structure and function level.	Cause described at the 4 levels using OMIM, UniProt, ClinVar and PDB.	Summary validated by the team and cited in Background.	14/10/2026	WP2
SO2	Compare wild-type and disease-associated SOD1 sequences.	2–3 pathogenic variants selected with explicit criteria; one alignment per variant; single residue-numbering convention.	Variant table + mapping table + alignments stored in the repository.	16/10/2026	WP2
SO3	Explain the structural and functional impact of the selected variants.	Every selected variant mapped onto the SOD1 3D structure and linked to protein function and patient phenotype.	Structure figures and healthy vs disease comparison table.	20/10/2026	WP2
SO4	Compare SOD1 between humans and at least two other species.	≥ 2 species selected and justified; MSA and conservation of the mutated residues analysed.	MSA figure and conservation analysis in Results.	19/10/2026	WP2
SO5	Communicate the findings in BMC Bioinformatics format and orally.	Report ≤ 5000 words / 7 pages and ≤ 5 figures/tables, delivered on GitHub; oral presentation by all members.	Final commit before 26/10 22:00; presentation on 05/11.	26/10/2026 · 05/11/2026	WP3, WP4

3. Work packages			
ID	Work package	Description	Main tasks
WP1	Project management & setup	Repository, planning (Gantt, risks), literature review and monitoring of the project status.	1.1–1.4
WP2	Bioinformatic analysis	Disease cause, variant selection, sequence alignments, 3D structure mapping and evolutionary conservation of SOD1.	2.1–2.3
WP3	Report writing	Writing, integration and review of the report following the BMC Bioinformatics template.	3.1–3.4
WP4	Oral presentation	Slides, cross-explanation of all sections and rehearsal for the oral defence.	4.1

4. Milestones				
ID	Milestone	Achieved when	Linked tasks	Date
M1	Gantt v1 delivered	Gantt, task details, milestones and risk analysis uploaded to Atenea (22:00) and frozen in gantt/.	1.2.3	09/10
M2	Variants and species selected	2–3 variants and ≥ 2 species chosen and justified; numbering mapping table ready.	2.1.2, 2.2.1	14/10
M3	Analyses completed	All alignments, structural mapping and conservation analyses finished.	2.2.3, 2.3.2	20/10
M4	Content freeze	All sections drafted; word count and figure limit checked. Only integration and review afterwards.	3.1.3, 3.3.1, 3.3.2, 3.4.1	21/10
M5	Status retrospective	Project Status reviewed with the lecturers.	1.4.3	22/10
M6	Final delivery	Last commit on GitHub before 22:00; later commits are discarded.	3.4.3, 3.4.4, 3.4.5	26/10
M7	Oral presentation	Project presented and defended by all members.	4.1.4	05/11

5. Project members, roles and responsibilities			
Member	Academic background	Role	Responsibilities
Ferran Sánchez	Bioinformatics bachelor's student (UPC-UB-UAB-UPF)	Project Manager & Structural Biology Analyst	Owns the repository; keeps the Project Status and live Gantt up to date and prepares the retrospective; maps the variants onto the 3D structure and analyses healthy vs disease differences; sets up the report skeleton and checks word count, figure limit, grammar and style.
Luz Hurtado	Bioinformatics bachelor's student (UPC-UB-UAB-UPF)	Product Owner & Disease Specialist	Accepts deliverables: submits the Gantt v1, checks references and the final delivery against the rubric and Formal Requirements; explains the disease cause and its impact on the patient; writes Background, the disease correlation and the Conclusions.
Noa Galván	Bioinformatics bachelor's student (UPC-UB-UAB-UPF)	Evolutionary Analysis Specialist & Report Integrator	Selects and justifies the comparison species, retrieves orthologs and runs the MSA, conservation and selective-pressure analyses; writes the evolutionary results and their interpretation; integrates all sections and writes the Abstract; structures the oral presentation.
Leyre Zazpe	Bioinformatics bachelor's student (UPC-UB-UAB-UPF)	Risk Manager & Sequence/Variant Analyst	Owns the risk register and monitors risks weekly; updates the repository to the Formal Requirements; selects the variants and builds the numbering mapping table; runs the wild-type vs mutant alignments; writes Methods and the mutant sequence results.

Progress (subtasks completed):

11%
Status updated up to: 09/10/2026

ID	Task	Owner	Plan. start	Plan. end	Real start	Real end	Days	Predecessor	Status	September 2026										October 2026										Nov 2026										
										14/09		21/09		28/09			05/10			12/10			19/10			26/10			02/11											
										17	18	21	22	23	24	25	28	29	30	1	2	5	6	7	8	9	12	13	14	15	16	19	20	21	22	23	26	27	28	29
1	Project management & setup		17/09	26/10			26																																	
1.1	GitHub environment setup		17/09	09/10			16																																	
1.1.1	Setting up repository	Noa & Ferran	17/09	17/09	17/09	17/09	1		Completed																															
1.1.2	Repository update to Formal Requirements (name, report/, references/, README)	Leyre	09/10	09/10	08/10	09/10	1	1.1.1	Completed																															
1.2	Project planning		01/10	09/10			7																																	
1.2.1	Gantt chart (Excel)	All	01/10	08/10	29/09	09/10	6	1.1.1	Completed																															
1.2.2	Risk assessment	Leyre	05/10	08/10	06/10	08/10	4	1.2.1 (SS)	Completed																															
1.2.3	Gantt v1 export to PDF, Atenea submission and freeze in repo	Luz	09/10	09/10			1	1.2.1, 1.2.2	Pending																															
1.3	Literature review		06/10	22/10			12																																	
1.3.1	Bibliography selection	All	06/10	09/10	06/10		4		In progress																															
1.3.2	Extraction of information and storage of sources	All	13/10	21/10			7	1.3.1	Pending																															
1.3.3	In-text citations and reference list (Oxford SCIMED)	All	15/10	22/10			6	1.3.1	Pending																															
1.4	Project monitoring		09/10	26/10			11																																	
1.4.1	Project Status and live Gantt updates (Mon & Thu)	Ferran	09/10	26/10			11	1.2.1	Pending																															
1.4.2	Risk monitoring	Leyre	09/10	26/10			11	1.2.2	Pending																															
1.4.3	Retrospective preparation	Ferran	21/10	21/10			1	1.4.1 (SS)	Pending																															
2	Bioinformatic analysis		13/10	20/10			6																																	
2.1	Sequence, mutation and structure analysis		13/10	16/10			4																																	
2.1.1	Disease cause at gene, protein, structure and function level	Luz	13/10	14/10			2	1.3.1	Pending																															
2.1.2	Variant selection and numbering mapping table	Leyre	13/10	14/10			2	1.3.1	Pending																															
2.1.3	Wild-type vs mutant sequence alignment	Leyre	15/10	16/10			2	2.1.2	Pending																															
2.1.4	Mapping of variants onto the 3D structure	Ferran	15/10	16/10			2	2.1.2	Pending																															
2.2	Evolutionary conservation		13/10	19/10			5																																	
2.2.1	Species selection (≥ 2) and ortholog retrieval	Noa	13/10	14/10			2	1.3.1	Pending																															
2.2.2	Multiple sequence alignment and conservation of mutated residues	Noa	15/10	16/10			2	2.1.2, 2.2.1	Pending																															
2.2.3	Selective pressure in evolution	Noa	19/10	19/10			1	2.2.2	Pending																															
2.3	Structure–function impact		19/10	20/10			2																																	
2.3.1	Healthy vs disease differences at gene, protein, structure and function level	Ferran	19/10	19/10			1	2.1.1, 2.1.3, 2.1.4	Pending																															
2.3.2	Impact on patient function	Luz	20/10	20/10			1	2.3.1	Pending																															

ID	Task	Owner	Plan. start	Plan. end	Real start	Real end	Days	Predecessor	Status	September 2026												October 2026												Nov 2026								
										14/09				21/09				28/09				05/10				12/10				19/10				26/10				02/11				
										17	18	21	22	23	24	25	28	29	30	1	2	5	6	7	8	9	12	13	14	15	16	19	20	21	22	23	26	27	28	29	30	2
3	Report writing (BMC Bioinformatics format)		13/10	23/10			9																																			
3.1	Background and Methods		13/10	21/10			7																																			
3.1.1	Report skeleton (BMC sections) and figure plan	Ferran	13/10	13/10	08/10		1	1.1.2	In progress																																	
3.1.2	Background	Luz	15/10	16/10			2	2.1.1, 3.1.1	Pending																																	
3.1.3	Methods	Leyre	21/10	21/10			1	2.1.3, 2.1.4, 2.2.2	Pending																																	
3.2	Results		19/10	20/10			2																																			
3.2.1	Mutant sequence findings	Leyre	19/10	20/10			2	2.1.3	Pending																																	
3.2.2	Evolutionary findings	Noa	20/10	20/10			1	2.2.3	Pending																																	
3.2.3	Structure and functional impact findings	Ferran	20/10	20/10			1	2.3.1	Pending																																	
3.3	Discussion and Conclusions		21/10	22/10			2																																			
3.3.1	Results interpretation	Noa	21/10	21/10			1	3.2.1, 3.2.2, 3.2.3	Pending																																	
3.3.2	Correlation with the disease	Luz	21/10	21/10			1	2.3.2, 3.2.1, 3.2.2, 3.2.3	Pending																																	
3.3.3	Conclusions	Luz	22/10	22/10			1	3.3.1, 3.3.2	Pending																																	
3.4	Integration and final review		21/10	23/10			3																																			
3.4.1	Word-count and figure-limit check	Ferran	21/10	21/10			1	3.2.1, 3.2.2, 3.2.3	Pending																																	
3.4.2	Integration of sections, Abstract and Keywords	Noa	22/10	22/10			1	3.1.3, 3.3.1, 3.3.2, 3.4.1	Pending																																	
3.4.3	Reference and citation check	Luz	23/10	23/10			1	1.3.3, 3.4.2	Pending																																	
3.4.4	Grammar and style check	Ferran	23/10	23/10			1	3.4.2	Pending																																	
3.4.5	Rubric and delivery checklist, Authors' contributions	Luz	23/10	23/10			1	3.3.3, 3.4.2	Pending																																	
4	Oral presentation		27/10	04/11			7																																			
4.1	Presentation preparation		27/10	04/11			7																																			
4.1.1	Slide structure and template	Noa	27/10	28/10			2	3.4.5	Pending																																	
4.1.2	Slides for own section	All	29/10	30/10			2	4.1.1	Pending																																	
4.1.3	Cross-explanation session	All	02/11	02/11			1	4.1.2	Pending																																	
4.1.4	Rehearsal with questions	All	03/11	04/11			2	4.1.3	Pending																																	
M	Milestones																																									
M1	Gantt v1 delivered (Atenea, 22:00)	—	09/10	09/10			0	1.2.3	Pending																																	
M2	Variants and species selected	—	14/10	14/10			0	2.1.2, 2.2.1	Pending																																	
M3	Analyses completed	—	20/10	20/10			0	2.2.3, 2.3.2	Pending																																	
M4	Content freeze	—	21/10	21/10			0	3.1.3, 3.3.1, 3.3.2	Pending																																	
M5	Status retrospective	—	22/10	22/10			0	1.4.3	Pending																																	
M6	Final delivery (GitHub, 22:00)	—	26/10	26/10			0	3.4.3, 3.4.4, 3.4.5	Pending																																	
M7	Oral presentation	—	05/11	05/11			0	4.1.4	Pending																																	

Legend

Completed — Task bar coloured by status (Status sheet)

In progress — Task bar coloured by status (Status sheet)

Incident — Task bar coloured by status (Status sheet)

Pending — Task bar coloured by status (Status sheet)

Canceled — Task bar coloured by status (Status sheet)

◆ Milestone — Checkpoint without duration

Holiday — Bank holiday (24/09 La Mercè, 12/10); weekends are not shown

Predecessors are finish-to-start unless marked (SS) = start-to-start.

Project Status — ALS1

Group 12DP-D (Ferran Sánchez · Luz Hurtado · Noa Galván · Leyre Zazpe) · Introduction to Bioinformatics 2026–2027 (UPC)

Updated up to: 09/10/2026												
Predecessors are finish-to-start unless marked (SS) = start-to-start (can start once the predecessor has started).												
ID	Task	Owner	Output	Accomplished when	Predecessor	Plan. start	Plan. finish	Duration	Real start	Real finish	Status	Notes
1	Project management & setup					17/09	26/10	26				
1.1	GitHub environment setup					17/09	09/10	16				
1.1.1	Setting up repository	Noa & Ferran	Shared GitHub repository with folder structure	All members are collaborators and can clone, commit and push		17/09	17/09	1	17/09	17/09	Completed	Repository created and folders added on 17/09
1.1.2	Repository update to Formal Requirements (name, report/, references/, README)	Leyre	Repository compliant with Formal Requirements	Repo named 12DP-D_ALS1_IB_2026; report/, status/, gantt/, references/ exist; README lists members, roles and structure	1.1.1	09/10	09/10	1	08/10	09/10	Completed	Remove unrelated entries (TTR, serum albumin) from the bibliography file
1.2	Project planning					01/10	09/10	7				
1.2.1	Gantt chart (Excel)	All	Gantt chart, task details and milestones	All tasks have owner, output, duration, predecessor and completion criteria	1.1.1	01/10	08/10	6	29/09	09/10	Completed	First Excel version committed on 29/09
1.2.2	Risk assessment	Leyre	Risk register and contingency plans	Each risk has score, category, contingency plan and owner; mandatory risk sentence written	1.2.1 (SS)	05/10	08/10	4	06/10	08/10	Completed	
1.2.3	Gantt v1 export to PDF, Atenea submission and freeze in repo	Luz	Gantt v1 PDF on Atenea and in gantt/	PDF uploaded before 20:00 on 09/10 and committed to gantt/	1.2.1, 1.2.2	09/10	09/10	1			Pending	Atenea deadline 22:00; upload at least 2 h earlier
1.3	Literature review					06/10	22/10	12				
1.3.1	Bibliography selection	All	List of relevant sources on ALS1 / SOD1	Key reviews and database entries (OMIM, UniProt, ClinVar, PDB) identified		06/10	09/10	4	06/10		In progress	Every member searches sources
1.3.2	Extraction of information and storage of sources	All	Notes per topic; source PDFs in references/	Each used source is stored in references/ (PDF preferred)	1.3.1	13/10	21/10	7			Pending	
1.3.3	In-text citations and reference list (Oxford SCIMED)	All	Citations in report.md and reference list	Every statement taken from a source is cited in Oxford SCIMED style	1.3.1	15/10	22/10	6			Pending	Done while writing each section
1.4	Project monitoring					09/10	26/10	11				
1.4.1	Project Status and live Gantt updates (Mon & Thu)	Ferran	Updated Status sheet, live Gantt and status/project_status.md	Real dates, status and notes updated every Monday and Thursday	1.2.1	09/10	26/10	11			Pending	Final Gantt must reflect real progress on 26/10
1.4.2	Risk monitoring	Leyre	Updated risk register	Score and status of each risk reviewed every week; activated contingencies recorded	1.2.2	09/10	26/10	11			Pending	
1.4.3	Retrospective preparation	Ferran	Status summary for the 22/10 retrospective	Delays, incidents and decisions documented in Notes	1.4.1 (SS)	21/10	21/10	1			Pending	
2	Bioinformatic analysis					13/10	20/10	6				
2.1	Sequence, mutation and structure analysis					13/10	16/10	4				
2.1.1	Disease cause at gene, protein, structure and function level	Luz	Summary of SOD1 gene, protein, structure and function in ALS1	Cause of ALS1 explained at the four levels with sources	1.3.1	13/10	14/10	2			Pending	
2.1.2	Variant selection and numbering mapping table	Leyre	2–3 selected variants + legacy ↔ HGVS mapping table	Variants chosen with explicit criteria (ClinVar classification, literature, structure availability); one numbering convention fixed	1.3.1	13/10	14/10	2			Pending	Contingency for risks 'too much information' and 'source mismatch'
2.1.3	Wild-type vs mutant sequence alignment	Leyre	Pairwise alignments and interpretation	Each selected variant located and described in the alignment	2.1.2	15/10	16/10	2			Pending	
2.1.4	Mapping of variants onto the 3D structure	Ferran	Structure figures with mutated residues highlighted	Each variant mapped onto the SOD1 structure (PDB) with its structural context	2.1.2	15/10	16/10	2			Pending	
2.2	Evolutionary conservation					13/10	19/10	5				
2.2.1	Species selection (≥ 2) and ortholog retrieval	Noa	Justified species list and SOD1 ortholog sequences	At least 2 species selected and justified; orthologs (not paralogs) retrieved	1.3.1	13/10	14/10	2			Pending	
2.2.2	Multiple sequence alignment and conservation of mutated residues	Noa	MSA and conservation analysis	Conservation of each selected residue described across species	2.1.2, 2.2.1	15/10	16/10	2			Pending	
2.2.3	Selective pressure in evolution	Noa	Interpretation of evolutionary constraint	Biological implication of conservation explained	2.2.2	19/10	19/10	1			Pending	
2.3	Structure–function impact					19/10	20/10	2				
2.3.1	Healthy vs disease differences at gene, protein, structure and function level	Ferran	Comparison table healthy vs disease	Differences described at the four levels for each variant	2.1.1, 2.1.3, 2.1.4	19/10	19/10	1			Pending	
2.3.2	Impact on patient function	Luz	Explanation of the physiological consequences	Link between molecular change and clinical phenotype explained with sources	2.3.1	20/10	20/10	1			Pending	

ID	Task	Owner	Output	Accomplished when	Predecessor	Plan. start	Plan. finish	Duration	Real start	Real finish	Status	Notes
3	Report writing (BMC Bioinformatics format)					13/10	23/10	9				
3.1	Background and Methods					13/10	21/10	7				
3.1.1	Report skeleton (BMC sections) and figure plan	Ferran	report/report.md skeleton and list of max. 5 figures/tables	All BMC sections present with owner and word budget	1.1.2	13/10	13/10	1	08/10		In progress	Brought forward: report structure uploaded by Leyre on 08/10. Section owners and word budget pending.
3.1.2	Background	Luz	Background section	Disease context, aims and research gap written and cited	2.1.1, 3.1.1	15/10	16/10	2			Pending	
3.1.3	Methods	Leyre	Methods section	Data sources, tools, versions and numbering convention described for reproducibility	2.1.3, 2.1.4, 2.2.2	21/10	21/10	1			Pending	
3.2	Results					19/10	20/10	2				
3.2.1	Mutant sequence findings	Leyre	Results subsection	Sequence findings written with figures/tables referenced	2.1.3	19/10	20/10	2			Pending	
3.2.2	Evolutionary findings	Noa	Results subsection	Conservation findings written with figures/tables referenced	2.2.3	20/10	20/10	1			Pending	
3.2.3	Structure and functional impact findings	Ferran	Results subsection	Structural findings written with figures/tables referenced	2.3.1	20/10	20/10	1			Pending	
3.3	Discussion and Conclusions					21/10	22/10	2				
3.3.1	Results interpretation	Noa	Discussion subsection	Findings compared with literature; limitations stated	3.2.1, 3.2.2, 3.2.3	21/10	21/10	1			Pending	
3.3.2	Correlation with the disease	Luz	Discussion subsection	Molecular findings linked to the ALS1 phenotype	2.3.2, 3.2.1, 3.2.2, 3.2.3	21/10	21/10	1			Pending	
3.3.3	Conclusions	Luz	Conclusions section	Main conclusions answer the project question	3.3.1, 3.3.2	22/10	22/10	1			Pending	Synthesis only: no new analyses after the 21/10 content freeze
3.4	Integration and final review					21/10	23/10	3				
3.4.1	Word-count and figure-limit check	Ferran	Checked draft	≤ 5000 words / 7 pages and ≤ 5 figures/tables	3.2.1, 3.2.2, 3.2.3	21/10	21/10	1			Pending	Contingency of the format risk (check on 21/10)
3.4.2	Integration of sections, Abstract and Keywords	Noa	Complete integrated report.md	Sections connected; structured Abstract ≤ 350 words; 5 keywords	3.1.3, 3.3.1, 3.3.2, 3.4.1	22/10	22/10	1			Pending	Synthesis only: no new analyses after the 21/10 content freeze
3.4.3	Reference and citation check	Luz	Verified reference list	All citations match the list and the PDFs in references/	1.3.3, 3.4.2	23/10	23/10	1			Pending	
3.4.4	Grammar and style check	Ferran	Proofread report	UK spelling (-ize), Oxford comma and number style applied	3.4.2	23/10	23/10	1			Pending	
3.4.5	Rubric and delivery checklist, Authors' contributions	Luz	Completed checklist and Declarations	All Formal Requirements and rubric items checked; contributions stated	3.3.3, 3.4.2	23/10	23/10	1			Pending	26/10 kept as buffer only (no new tasks)
4	Oral presentation					27/10	04/11	7				
4.1	Presentation preparation					27/10	04/11	7				
4.1.1	Slide structure and template	Noa	Slide template and outline	Outline agreed by the team	3.4.5	27/10	28/10	2			Pending	After final delivery; nothing committed after 26/10 22:00
4.1.2	Slides for own section	All	Complete slide deck	Each member's slides added to the deck	4.1.1	29/10	30/10	2			Pending	Each member prepares the part they present
4.1.3	Cross-explanation session	All	Shared understanding of all sections	Each member can explain every section (risk: niche awareness)	4.1.2	02/11	02/11	1			Pending	
4.1.4	Rehearsal with questions	All	Rehearsed presentation	Full run within time; questions on other members' sections answered	4.1.3	03/11	04/11	2			Pending	
M	Milestones											
M1	Gantt v1 delivered (Atenea, 22:00)	—			1.2.3	09/10	09/10	0			Pending	
M2	Variants and species selected	—			2.1.2, 2.2.1	14/10	14/10	0			Pending	
M3	Analyses completed	—			2.2.3, 2.3.2	20/10	20/10	0			Pending	
M4	Content freeze	—			3.1.3, 3.3.1, 3.3.2	21/10	21/10	0			Pending	
M5	Status retrospective	—			1.4.3	22/10	22/10	0			Pending	
M6	Final delivery (GitHub, 22:00)	—			3.4.3, 3.4.4, 3.4.5	26/10	26/10	0			Pending	
M7	Oral presentation	—			4.1.4	05/11	05/11	0			Pending	

Risk management — ALS1

Group 12DP-D (Ferran Sánchez · Luz Hurtado · Noa Galván · Leyre Zazpe) · Introduction to Bioinformatics 2026–2027 (UPC)

ID	Risk	Type	L	I	S	Category	We have realized because...	Contingency plan	Related task	Owner	Status
R1	Using terms we don't understand or including GenAI errors or hallucinations	Scientific	2	3	6	High	SOD1 literature uses highly specialised terminology and the IB GenAI policy makes us liable for any GenAI-generated content.	Each section is cross-read by the other members, who must be able to understand and explain it. GenAI is only used for wording and formatting, outside GitHub, and its output is always checked manually.	3.4.3	Luz	Open
R2	Tight schedule: deadline	Team & planning	2	3	6	High	The final deadline on Atenea is Monday 26/10 at 22:00, and only two working days remain after the 22/10 retrospective.	Content freeze on 21/10; 22–23/10 for integration and review only; 26/10 kept exclusively as a buffer.	1.4.1	Ferran	Open
R3	Uneven contribution	Team & planning	2	3	6	High	Our first Gantt draft assigned several tasks to "All"	Every task has a single owner and a balanced workload; each member commits their own work at least twice a week. Tasks are reassigned early if someone falls behind.	1.4.1	Ferran	Open
R4	"Niche awareness" at the oral presentation	Team & planning	2	3	6	High	Our analysis tasks are split by specialisation, each owned by a single member, while the rubric assesses every member's understanding of the whole project.	Short weekly sync where each member explains their progress to the rest; each section is cross-read by a different member; everyone reads the full report before the presentation; a rehearsal with questions about other members' sections.	4.1.1	Noa	Open
R5	Mismatch between different sources leads to wrong positions in alignments and structures	Scientific	3	2	6	High	The same SOD1 mutation is named A4V in the classic literature but p.Ala5Val in ClinVar.	Before any analysis, define a single numbering convention in Methods and build a mapping table between sources.	2.1.2	Leyre	Open
R6	Trying to analyse too much information (e.g. too many variants)	Scientific	3	2	6	High	ClinVar currently lists 194 SOD1 variants classified as pathogenic or likely pathogenic (consulted 08/10), far more than we can analyse.	Select 2–3 representative variants using explicit criteria by a fixed date (14/10); anything outside that scope is left for the Discussion as future work.	2.1.2	Leyre	Open
R7	Format errors in the submissions	Technical & format	2	2	4	Medium	The Formal Requirements and the BMC template impose many formal rules	Build a checklist from the Formal Requirements and the BMC template (repo name, mandatory folders, .md report, word limits...) and review it before each delivery. Agree an approximate word budget per section from the start. Check the word count on 21/10.	3.4.5	Luz	Open
R8	Technical problems with Git files or GitHub	Technical & format	2	2	4	Medium	Our repository did not meet some formal rules yet (it was named ALS1 and had no report/ folder)	Small and frequent commits. Upload deliverables at least 2 h before the deadline.	1.1.2	Leyre	Open
R9	Gantt and Status not kept up to date	Team & planning	2	2	4	Medium	Our first versions of the Gantt and of the Project Status had empty fields and outdated dates.	One owner updates the Project Status and the live Gantt every Monday and Thursday with real dates, status and notes; each member reports their progress at each meeting.	1.4.1	Ferran	Open
R10	Distant species make the comparison hard to interpret	Scientific	1	2	2	Low	The project requires comparing human SOD1 with at least two other species, and the candidates range from close mammals to very distant organisms with much less similar sequences.	Choose species with a clear biological or experimental justification. If the distant species alignment is not informative, replace it with an intermediate one and explain the choice in Methods.	2.2.1	Noa	Open
R11	Online tools unavailable or slow close to the deadline	Technical & format	1	2	2	Low	The tools we need (BLAST, multiple sequence alignment servers) are shared public services with job queues, used by all groups in the same weeks.	Run all analyses early, well before the content freeze; save every output (files + screenshots, with date and tool version).	2.2.2	Noa	Open